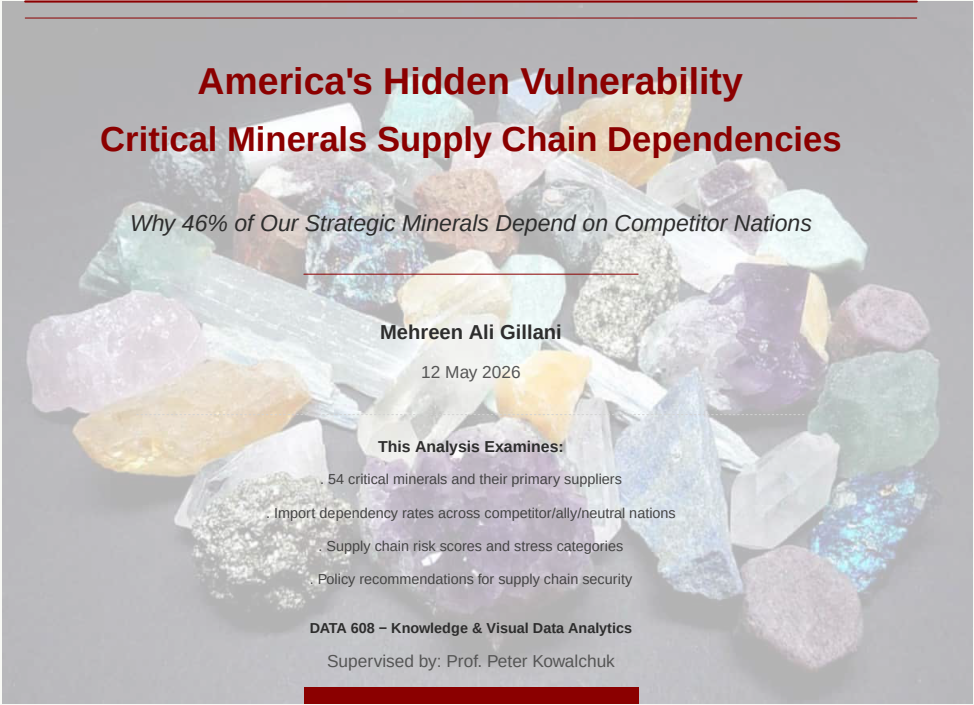




America's Hidden Vulnerability Critical Minerals Supply Chain Dependencies

Why 46% of Our Strategic Minerals Depend on Competitor Nations

Mehreen Ali Gillani

12 May 2026

This Analysis Examines:

- . 54 critical minerals and their primary suppliers
- . Import dependency rates across competitor/ally/neutral nations
- . Supply chain risk scores and stress categories
- . Policy recommendations for supply chain security

DATA 608 – Knowledge & Visual Data Analytics

Supervised by: Prof. Peter Kowalchuk

Data: USGS Mineral Commodity Summaries 2023–2024

The Strategic Reality

Critical Minerals: The Foundation of Modern America

WHAT IF SUPPLY FAILS?

What Are They?

Critical Minerals are essential to:

- . National Defense (radar, guided missiles)
- . Clean Energy (EV batteries, solar panels)
- . Technology (smartphones, semiconductors)
- . Medical Devices (MRI machines, implants)

STRATEGIC VULNERABILITY

The China Factor

**China supplies 23 of the 54
critical minerals (46%)**

Dominance in:

- . Rare Earth Elements (100% import)
 - . Battery metals (Lithium, Cobalt)
- . Semiconductor materials (Gallium)

No other nation has this level of concentrated c

THE GLOBAL SUPPLY LANDSCAPE

Allies, Neutrals & Competitors

TOP SUPPLIER NATIONS FOR US CRITICAL MINERALS

By Country Alignment Category

COMPETITOR NATIONS

1. China
→ 25 minerals (46%)
→ 18 at Critical stress
2. Russia
→ Vanadium

ALLY NATIONS

1. Canada
2. Norway
3. Australia
4. Japan
5. Germany
6. Mexico
7. United Kingdom

NEUTRAL NATIONS

1. South Africa
2. Brazil
3. Chile
4. India
5. Kazakhstan

KEY TAKEAWAY

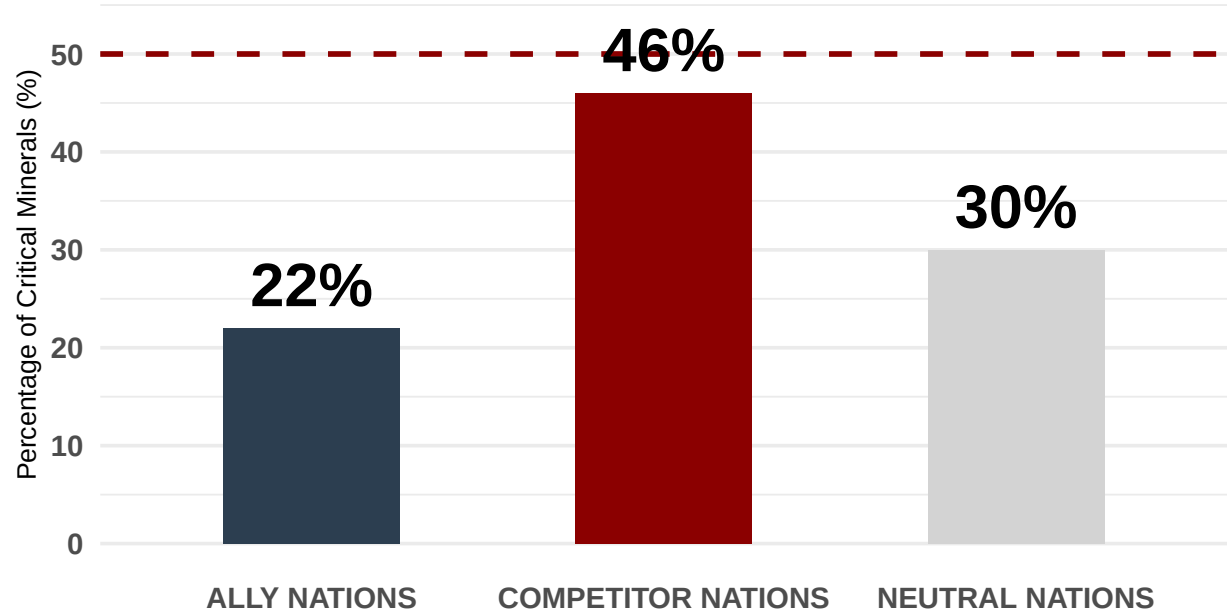
1 nation (China) supplies 46% of all US critical minerals
No other single nation poses the same level of supply chain risk

The Core Truth

America’s Critical Minerals Pipeline Runs Through Adversaries

ALMOST HALF OF OUR CRITICAL MINERALS COME FROM COMPETIT

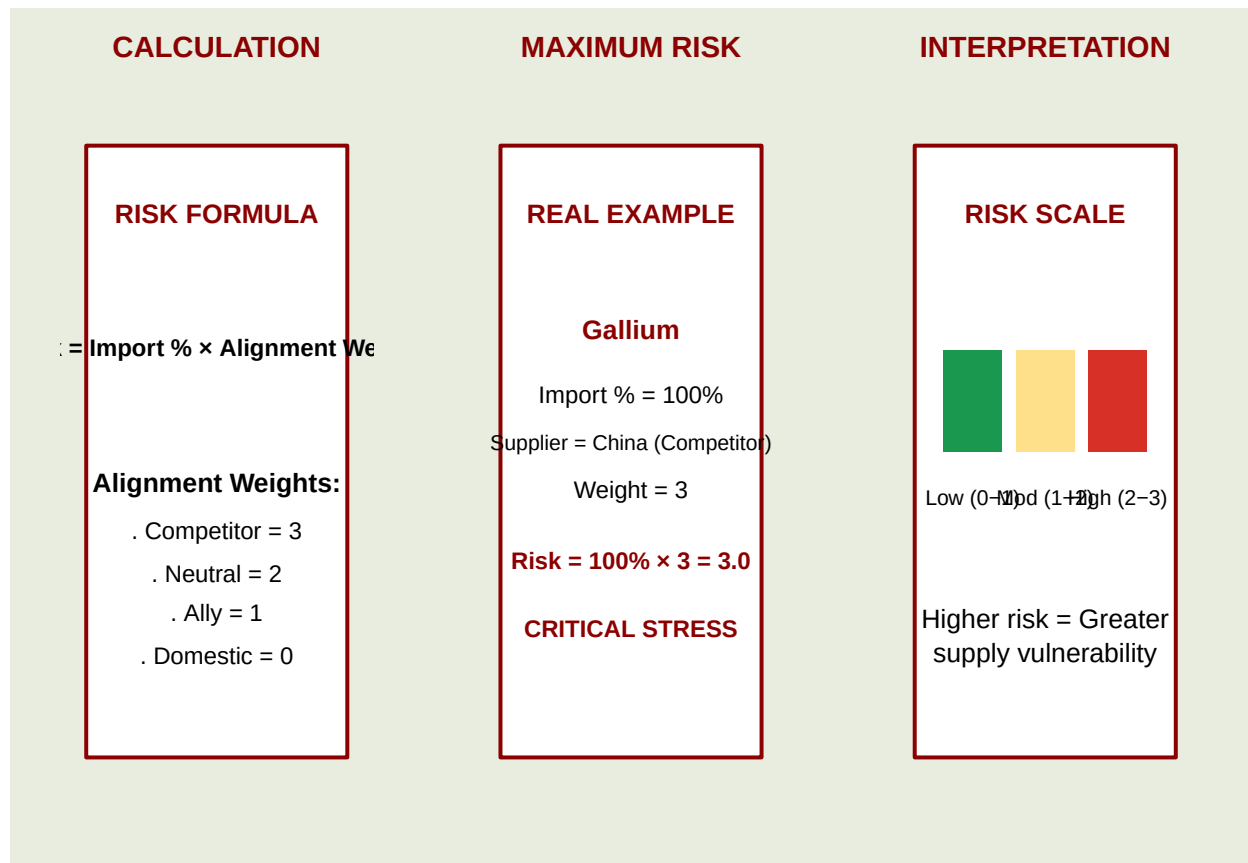
Only 22% come from allied nations | Competitor threshold: 46%



Competitor nations include China and Russia

The Risk Score Methodology

How We Calculate Supply Chain Vulnerability

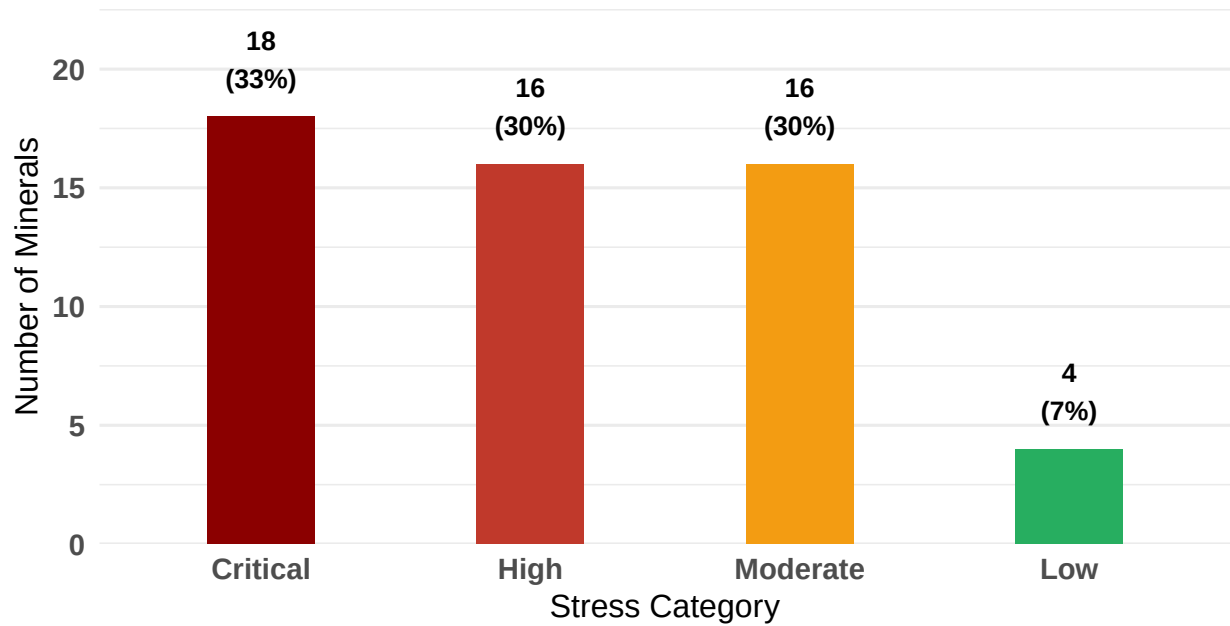


The Supply Stress Categories

34 Minerals (63%) at High or Critical Risk

SUPPLY STRESS LEVELS: 63% AT HIGH OR CRITICAL

Critical = 100% import from competitors | High = $\geq 75\%$ import from Competitor/Neutral



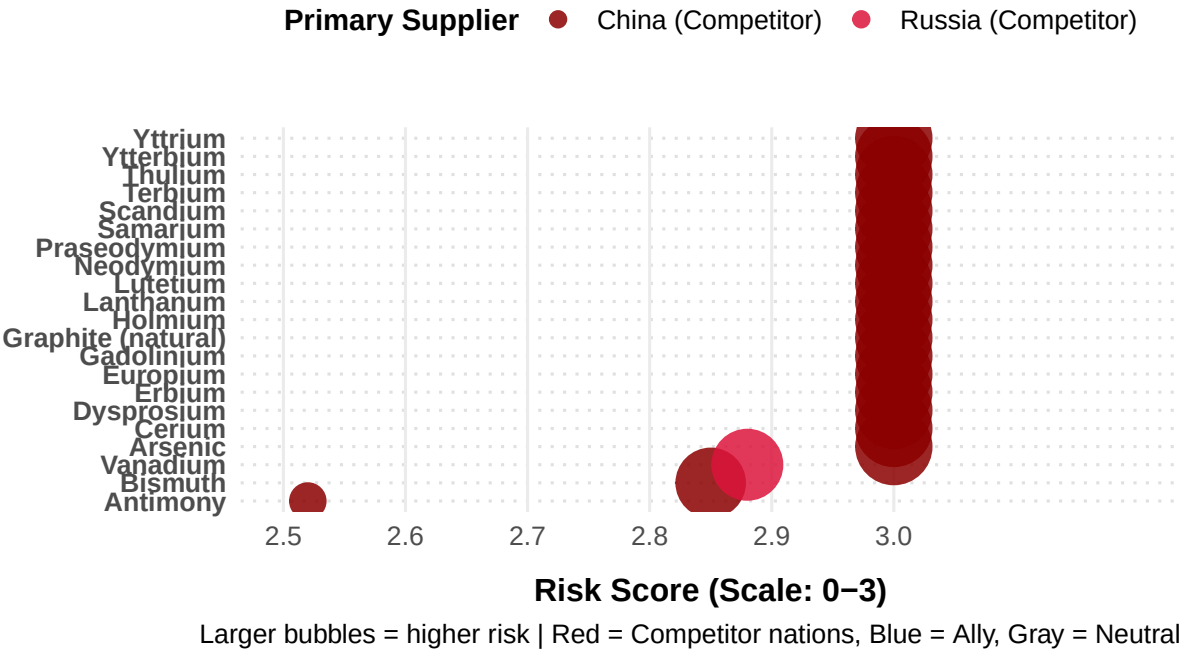
Critical minerals requiring immediate attention: 18

The Highest Risk Minerals

18 Critical-Stress Minerals (100% from Competitors)

CRITICAL STRESS MINERALS

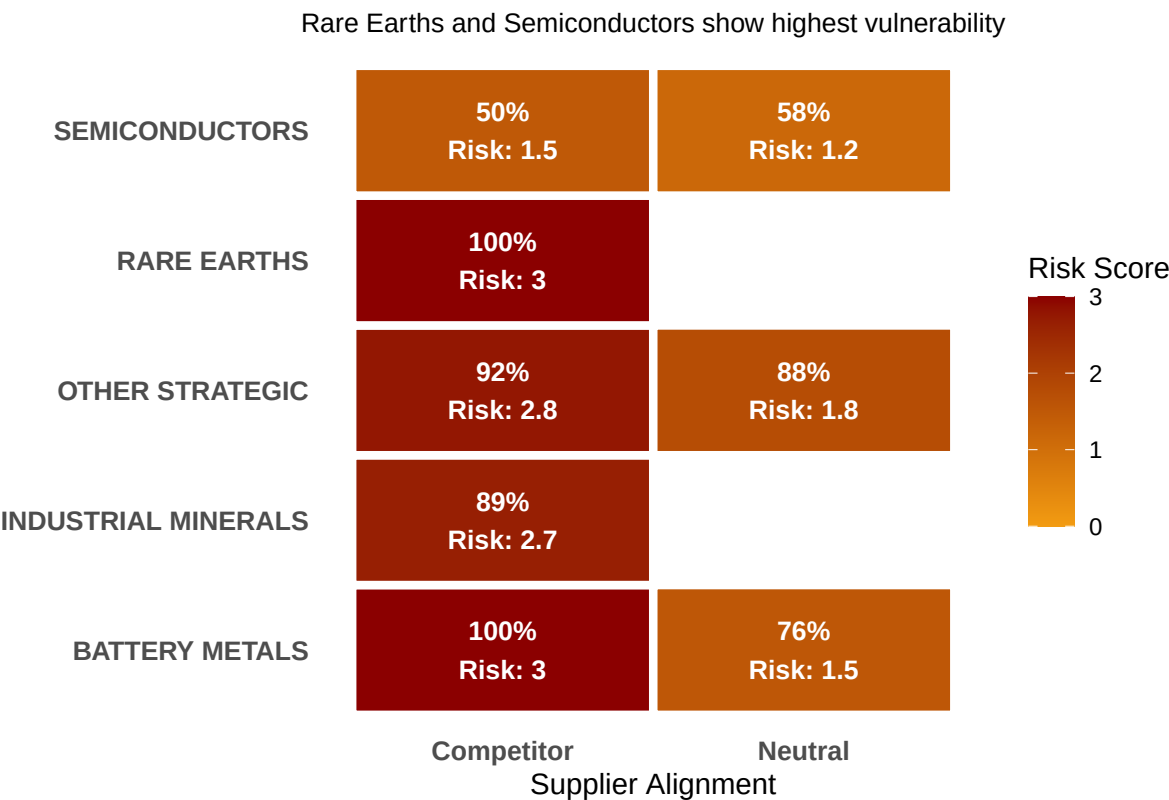
Risk scores near maximum (3.0) | 100% import rate for all



The Competitor & Neutral Supplier Heatmap

Which Mineral Categories Face the Greatest Threat?

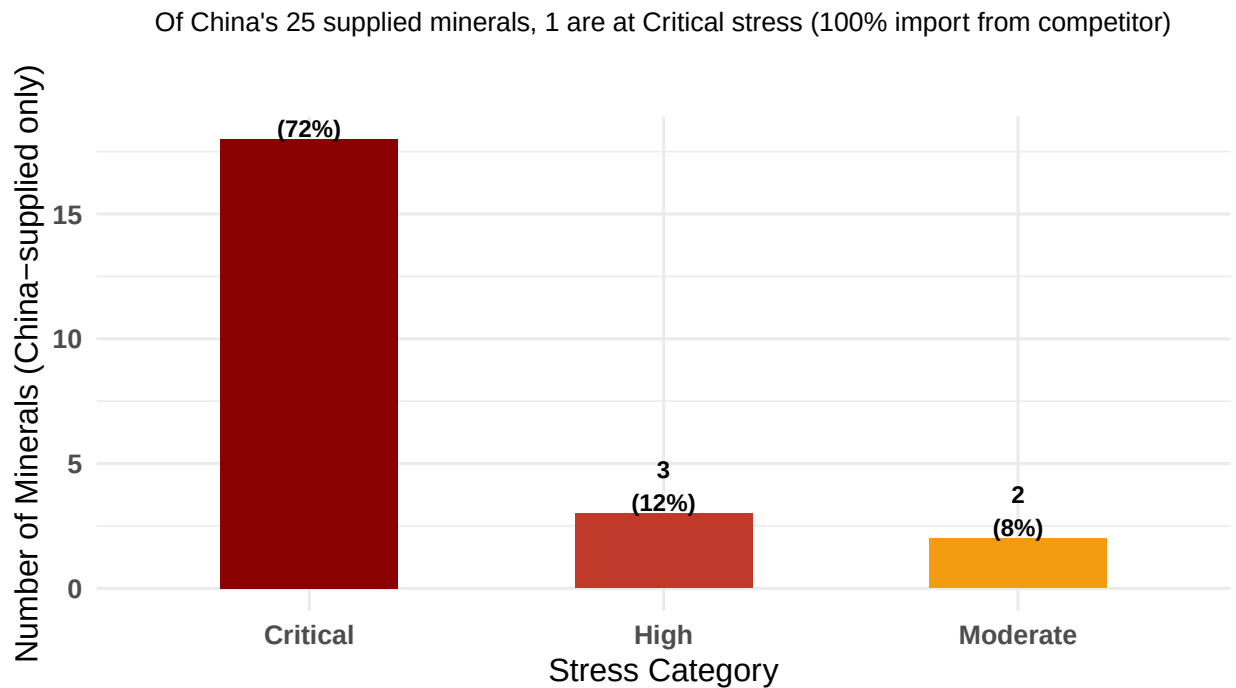
COMPETITOR & NEUTRAL SUPPLY RISK BY MINERAL CATEGORY



The China Factor

One Nation Controls Our Most Critical Supply Chains

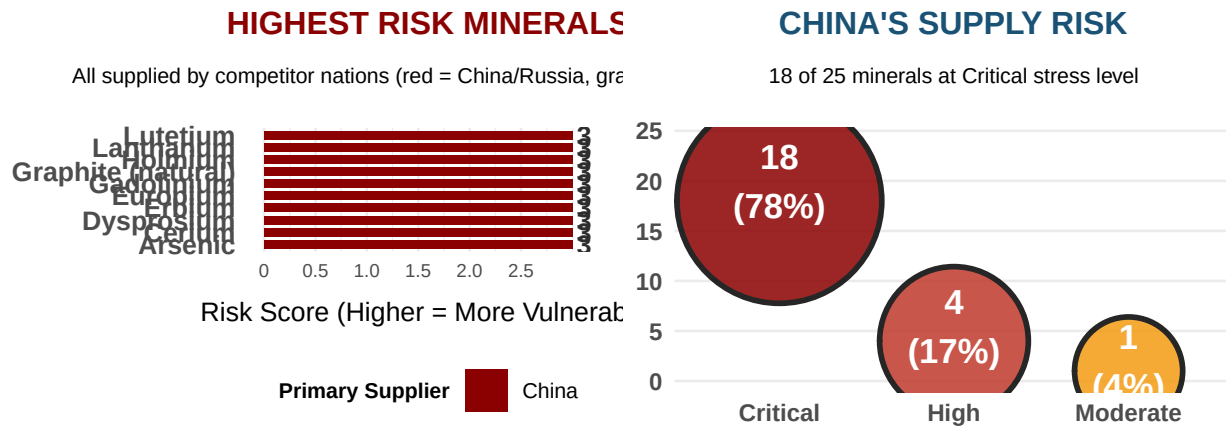
CHINA'S GRIP: 25 MINERALS, 18 AT CRITICAL STRESS



China is the dominant supplier for Rare Earths, Battery Metals, and Semiconductors

Two Vulnerabilities: Rates vs Volume

High-Risk Minerals vs High-Volume Supply Lines



KEY INSIGHT: China dominates high-risk supply chains

- . 18 minerals at CRITICAL stress (100% import from competitors)
- . 25 total minerals supplied by China
- . Immediate diversification needed for Critical-stress minerals

Key Findings: What The Data Tells Policymakers

Four Unavoidable Truths About Critical Mineral Security



Strategic Recommendations

Three-Tier Response Based on Risk Level

Evidence-Based Policy Response by Risk Category

TIMELINE

TARGET

ACTION

EXPECTED IMPACT

IMMEDIATE

Critical-Stress Minerals (18)

Expand National Defense Stockpile Emergency diversification sourcing Diplomatic engagement with allies

6-month supply buffer Reduce immediate vulnerability

SHORT-TERM (6-12 mo)

High-Risk Minerals (16)

Domestic mining permits & incentives Recycling infrastructure investment R&D for substitutes

Reduce import dependency 25-40% Create circular economy

LONG-TERM (2-5 yr)

All Strategic Minerals

Build allied supply chains Trade agreements with neutral nations Education & workforce development

Long-term supply security Competitive domestic industry

THANK YOU

Questions & Discussion

Data Source:
USGS Mineral Commodity

Analysis:
54 critical minerals analyzed | Risk assessment across 3 alignment categories

Contact:
Mehreen Ali Gillani | DATA 608 – Knowledge & Visual Data Analytics

*"Our critical minerals supply chain is a national security issue.
With 46% coming from competitor nations and 18 minerals at critical stress,
the time for strategic diversification and allied partnerships is now."*

Supervised by: Prof. Peter Kowalchuk